

Enhancement One: Software Design and Engineering Narrative

The artifact I chose to represent software design and engineering is my android inventory tracking application created in the java programming language. It was designed to track inventory items in an SQLite database where the user can access inventory items to either read items on a grid view, add new items by name and quantity, update item by quantity, or delete an item. I created this application at the end of 2025 for my mobile architect and programming course at Southern New Hampshire University.

I chose this artifact to showcase my skills and ability in software development because it demonstrates my ability to work with the java programming language that incorporates object-oriented principles. It is structured with Model View Controller (MVC) architecture that is appropriate to Android development. My model layer is the DatabaseHelper.java where it manages the SQLite database and handles the CRUD operations. The RecyclerView_list.java is the model class to represent one inventory item. The view is the xml layout files, and the controller is the Activity.java files handling the user interactions and connect the view. This demonstrates competency in building industry standard coding practices that make the application readable, reusable, maintainable, and modular. Working in Android Studios shows efficiency in using an IDE for development and debugging skills. In this project I was able to incorporate UI/UX design principles with the use of XML layouts that incorporate material design practices that are appropriate to android development. Providing me with a foundational knowledge base on how to build software and implement it with the user in mind.

The artifact was improved by writing self-documented comments, refactoring naming conventions for the user and item tables that made the code easier to read and maintain. Constants were added to the columns of the items table to implement defensive programming.

Branching of my if conditions were simplified to return Boolean values. The removal of all SMS permission files to eliminate security vulnerabilities and complexity of the program of a feature that was no longer needed. A method was created to check if an item already existed in the database to improve programming logic, provide input validation, and support the integrity of data by preventing duplication of items. This also enhanced the user interface by preventing an overflow of repeated items on the grid view.

I have completed all planned outcomes for this enhancement. My primary files within this enhancement were the DatabaseHelper.java and the ActivityAddItem.java. This is a cohesive artifact, meaning all project files will be used for the final project, but I am slowly working through the files that align with the code review in areas such as structure, documentation, variable consistency and naming, conditional branches, and defensive programming. I will demonstrate course outcome numbers two, three, and five represented in the programming files and the narrative document.

Some of the challenges I faced when modifying my artifact was simply remembering and understanding my own code. I have all the references from when I created the project, but when I went into the files I did not implement self-documented code efficiently and in some areas there was no comments at all. I had to do quite a bit of research to remember why I wrote the queries I did, why I used selectionArgs, and so on. It essentially felt like I was relearning and reteaching my own program. I was able to fix my documentation and rename and restructure my code where it would be understandable to my future self and realistically to my professor. This process was beneficial in terms of recognizing other potential flaws in my program, for instance with larger databases utilizing a cursor to iterate through each row, may not be a practical approach, however this is something I can explore further in enhancement three. I have also found new and

helpful resources to teach me more about coding best practices and how to implement safer code.